Radial Basis Function (RBF) Kernel

Example-3

$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$

Normal Model

$\begin{pmatrix} u_1 & u_2 & v_1 \\ & & v_2 \end{pmatrix}$

Map Input data from 2-D to infinite Dimension

$\begin{pmatrix} u_1 & v_1 \\ u_2 & v_2 \end{pmatrix}$

$e^{-\gamma ||u - v||}$

Vectors Mapping

$\vec{a}, \vec{b} \rightarrow \vec{u}, \vec{v}$

Higher Computational Complexity !!!

$\begin{matrix} u_1 & v_1 \\ u_2 & v_2 \\ u_3 & v_3 \\ u_{\dots} & v_{\dots} \\ u_{\dots} & v_{\dots} \\ u_{\infty} & v_{\infty} \end{matrix}$

Kernelized Model

$\begin{pmatrix} u_1 & u_2 & u_3 & \dots & \dots & u_{\infty} & v_1 \\ & & & & & & v_2 \\ & & & & & & v_3 \\ & & & & & & v_{\dots} \\ & & & & & & v_{\dots} \\ & & & & & & v_{\infty} \end{pmatrix}$

Higher Computational Complexity !!!